

PracticalBP: Continuous Cuffless Blood Pressure Monitoring with Only One Record for Calibration

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Frequent blood pressure (BP) monitoring is crucial for helping patients and clinicians assess the risk of cardiovascular disease. The current oscillometric method needs to occlude peripheral arteries and the repeated inflation can cause discomfort, leading to reduced compliance. Although several cuffless BP monitoring systems have been proposed, they often either lack sufficient accuracy or impose a significant calibration burden, rendering them impractical. To address this gap, we introduce PracticalBP, a cuffless wrist-worn system that requires only a single BP reading for calibration to achieve accurate BP measurements. We identify the main issue of current cuffless devices is that the key user-specific hemodynamic information cannot be obtained with a low effort. Our key insight is that these individual BP-related information can be obtained effortlessly by physiology-based methods. To facilitate this, we propose a physiology-based pseudo-label generation mechanism to create an individual data distribution based on the principles of tonometry and hydrostatic pressure in arteries. During it, the user only needs to simply stay calm and tilt the arm. Furthermore, we observe that BP-related features exhibit different patterns across individuals. Therefore, we propose an effective feature extraction approach that combines a general feature extractor with a feature-wise attention module to assess the importance of each feature for different subjects. Additionally, in order to improve robustness, we design a channel-based data augmentation mechanism to cope with the signal distortion caused by the sensor position changes during each wear. PracticalBP is implemented in the real system with our developed wristband. We collect a dataset from 25 healthy subjects and 18 hypertension patients and conduct both rest tests and BP intervention experiments. The evaluation demonstrates that PracticalBP achieves an overall systolic BP (SBP) and diastolic BP (DBP) estimated error (mean $\pm$ standard deviation) of (-1.49 ± 7.89) mmHg and (-0.83 ± 6.23) mmHg, respectively, which meets the requirement of AAMI standard. Notably, our system achieves comparable accuracy to state-of-the-art solutions while reducing the calibration time by a factor of five and eliminating the need for complex BP intervention tasks.

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2474-9567/2025/9-ART100

<https://doi.org/10.1145/3749486>

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Applied computing** → **Consumer health**; • **Hardware** → **Sensor applications and deployments**.

Additional Key Words and Phrases: Cuffless blood pressure monitoring, Wearable device, Neural network, Hypertension

ACM Reference Format:

Chenggao Li, Junyao Peng, Qingyong Hu, Lin Chen, Yandao Huang, Shun Wu, Weibin Cheng, and Qian Zhang. 2025. PracticalBP: Continuous Cuffless Blood Pressure Monitoring with Only One Record for Calibration. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 3, Article 100 (September 2025), 31 pages. <https://doi.org/10.1145/3749486>

1 INTRODUCTION

Hypertension is a major cause of premature death worldwide. According to a WHO report, around 1.28 billion adults aged 30–79 years worldwide have hypertension. High blood pressure (BP) usually has no obvious symptoms but it's a major risk factor for stroke, heart failure, and dementia [30, 43]. What's worse is that the well-known white coat hypertension (WCHT) cannot be diagnosed in the hospital while the subjects are present in about 20% of people who appear to have hypertension [27]. Ambulatory BP measurement (ABPM) which requires the user to take BP measurements every 15 or 30 minutes in a day, is the gold standard to diagnose WCHT. ABPM is also crucial for abnormal BP assessment and BP management. The oscillometric method is the most commonly adopted way to measure BP. However, it necessitates the use of an inflatable cuff that compresses the arm's arteries until they are occluded. This process can cause discomfort and, in some cases, injury during frequent BP measurements [46, 68].

In recent years, cuffless BP monitoring has attracted the attention of many researchers. Some prior works tried to build a mechanism-based model [11, 12, 54] to bridge the physiological feature (e.g., pulse transit time) and BP. However these simplified models are based on some ideal assumptions, such as the constant thickness of blood vessels, and lead to unsatisfactory accuracy and frequent calibration. In addition, thanks to the powerful fitting ability of deep learning models, some researchers applied neural network to learn BP-related features automatically from electrocardiogram (ECG) [16], photoplethysmography (PPG) [7, 61, 65, 70], ballistocardiogram (BCG) [62] or pulse pressure waveform (PPW) [35, 36]. The models used in these works can be divided into two categories, general model and individual information fused model. The first type of works designed advanced deep learning models to extract general features [35, 36, 61, 62, 65, 70]. However, since BP is the result of multiple complex user-specific cardiovascular parameters, they perform poorly in large-scale datasets especially containing hypertensive patients.

Consequently, other studies have sought to combine individual parameters [7, 74] to enhance the system's practicality. *BP³* [74] incorporates users' physiological factors, such as gender, age, BMI, and hypertension history, and employs a physio-pulse attention mechanism to integrate these factors with pulse data collected via mmWave technology. While leveraging statistical parameters improves accuracy, the performance of this approach can degrade significantly in certain unaccounted scenarios, such as during medication use. In contrast, *FewShotBP* [16] trains a personalization adapter for each user using an individual dataset. This dataset comprises 10 cuff-based BP records obtained during rest, cold pressor tests, and stair tests. By utilizing actual labels, *FewShotBP* can directly learn the user-specific mapping of features in the input signals to BP values. However, the requirement to collect personal datasets through time-consuming (at least 20 minutes) and inconvenient tasks renders *FewShotBP* impractical. What's more, the 10 oscillometric BP measurements in *FewshotBP* may lead to arm soreness in users due to blocked blood circulation caused by repeated inflation of the cuff [18], thereby reducing user compliance. To the best of our knowledge, no cuffless BP monitoring system currently achieves high accuracy with practical calibration efforts.

To bridge this gap, we propose PracticalBP, an accurate continuous cuffless BP monitoring system with only one cuff-based BP record for calibration. Our key observation is that individual BP-related information can be

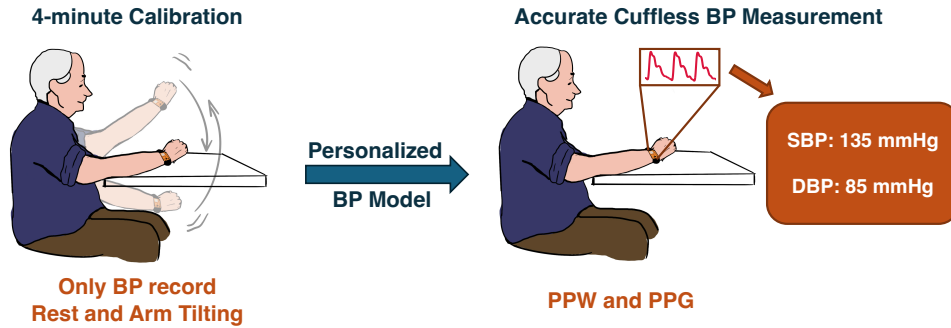


Fig. 1. Usage scenario.

introduced by simple physiological principles. Specifically, we leverage the arterial applanation tonometry and hydrostatic pressure in the artery to capture the user's own hemodynamic information that is cardiac output (CO) and total peripheral resistance (PR) information, respectively. CO and PR are considered to be the two most important factors affecting BP. Additionally, these two physiological methods can be conducted with a low effort. With only one actual BP reading, tonometry can track the short-term arterial BP variation by measuring the PPW above the radial artery at rest (Section 2.2). Hydrostatic pressure is the gravity-induced BP difference at different locations of the artery. Various local BP records can be obtained by simple arm tilting due to the fixed mapping between relative height and BP changes (Section 2.3). We will further explain the rationality of the pseudo labels we generated in Section 3. To this end, we design a wristband that includes a pressure sensor array, an IMU, and a PPG sensor. The pressure sensor array ensures the PPW at the thin radial artery can be easily collected. The collected radial PPW contains abundant BP-related information (Section 2.4) and is also used in the BP inference model. The IMU is used to sense the changes in height for hydrostatic pressure measurement. In addition, we deploy a PPG sensor that can reflect changes in arterial blood volume upstream of the radial artery for BP-related information supplementation. Figure 1 shows the use of PracticalBP. When a new user comes, the user will first wear our customized wristband and measure BP once with a cuff-based device. Then the user will be asked to keep still and title arm in turn for building a personalized BP inference model. The whole procedure of calibration will be finished within four minutes in a sitting position. After that, the user can measure BP in an ultra-low delay with collected PPW and PPG signals.

To transform this high-level idea into practice, we need to address three major challenges as follows:

1) *How to capture the accurate user-specific hemodynamic information in practical?*

We use a pressure sensor array to operate tonometry. To track the BP variation precisely, finding the sensor placed above the radial artery is necessary. Although we can choose the channel with the largest pulse strength, the multiple linear regions and inconsistent sensitivity of film sensors will cause wrong selection. For the measurement of hydrostatic pressure, an IMU-based method can be used. However, the IMU-based displacement measurement suffers from cumulative errors caused by the inherent drift. These two problems will make the obtained user-specific hemodynamics information inaccurate.

2) *How to extract BP-related features effectively from input signals and map them to BP?*

Due to the different hemodynamic parameters of each person, the key features extracted from PPW and PPG are different for each person. For example, the tidal peak, which is highly correlated with BP, is missing in some people's signals, and the waveforms of hypertensive patients are also significantly different from those of healthy people. An effective feature extractor is needed to improve the generalizability of the system.

3) *How to improve system robustness towards sensitive PPW under the sensor position deviation?*

The PPW is susceptible to the sensor position. The discrete sensor distribution in the array makes it possible for the target artery to lie between adjacent sensors. In addition, each time the user wears our device, the relative position of the sensor to the radial artery will change. These situations will both lead to unstable features.

To tackle these challenges, we propose a set of designs accordingly:

- 1) **Physiology-based Pseudo Label Generation.** We design a waveform-based channel selection algorithm to improve the labeling accuracy of the tonometry. It can relieve the impact of the multi-linear interval of the sensor while maintaining the stability of the feature to the greatest extent. With regard to hydrostatic pressure, considering that we only need to know the height of a few discrete positions, we design a mechanism that only focuses on the displacement of the forearm dynamic process to effectively reduce the error of IMU drift. With the help of them, we can derive many BP labels. In this paper, we call these BP labels physiology-based pseudo labels.
- 2) **Effective Feature Extraction.** This mechanism consists of two modules. First, we extract general BP-related features with both data-driven and manual approaches. These features are based on large-scale datasets and experience. Then, at the model personalization stage, we propose a feature-wise attention module to determine the importance of all generic BP-related features to a specific user with the personal dataset. Moreover, to fuse modalities of PPW and PPG, another cross-attention module that can leverage the deep mapping correlations between the two modal features is used.
- 3) **Channel-based Data Augmentation.** Due to the inconsistent sensitivity of the sensors in the array, we are unable to reconstruct PPW in the accurate position based on multi-channel PPW. Instead, we focus on the features that remain consistent across different sensor locations. To this end, we augment our dataset by constructing the PPW around the best channel in the array and enforcing the model to learn the features that are insensitive to the sensor position. With this mechanism, the robustness of our system is improved.

We implement PracticalBP on a customized wristband that carries a 3×3 piezoresistive sensor array, a PPG sensor, and an IMU. To adequately evaluate it, we collect a real-world dataset of 43 volunteers that contains 25 healthy subjects and 18 hypertensive patients. In addition, except for the resting state, we conduct BP intervention experiments including cold pressor and breath holding in the healthy group. These two tests can temporarily increase and decrease the BP of the subjects respectively. Under the leave-one-subject-out (LOSO) setting, the results show that PracticalBP achieves a low systolic BP (SBP) and diastolic BP (DBP) estimated error (mean ± standard deviation) of (-1.49 ± 7.89) mmHg and (-0.83 ± 6.23) mmHg, respectively, which are within the AAMI [66] requirement of (5 ± 8) mmHg. The performance is comparable to the state-of-the-art (SOTA) [16], while the time cost of calibration is reduced five times, and inconvenient BP intervention tasks are exempted. Our key contributions are concluded as follows:

- We propose PracticalBP, a continuous cuffless blood pressure monitoring system that achieves comparable accuracy to SOTA with only one BP record for calibration. To the best of our knowledge, the calibration effort of our system is the lowest among the existing works.
- We design a set of techniques, including a waveform-based channel selection algorithm and a discrete height measurement that can help to generate accurate personalized pseudo BP labels. In addition, we propose a hybrid BP-related feature extractor and a feature-wise attention mechanism for effective feature extraction. Furthermore, we design a channel-based data augmentation method to improve the system robustness to sensor location deviation.
- We prototype PracticalBP with a customized wristband. Extensive experiments on 43 subjects, including healthy subjects and hypertensive patients, demonstrate an excellent performance in cuffless BP measurement.

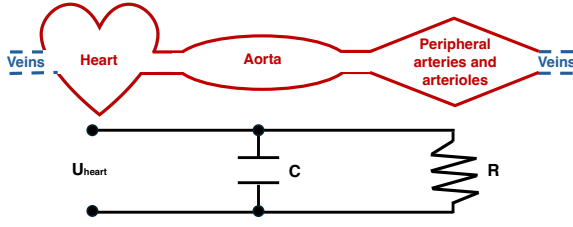


Fig. 2. Windkessel model.

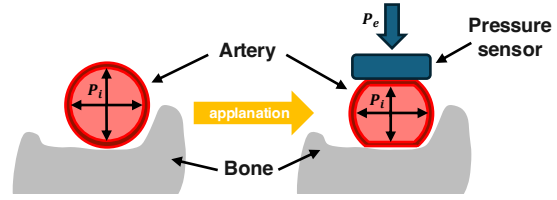


Fig. 3. Arterial applanation tonometry.

2 PRELIMINARIES

2.1 Influence Factors on Blood Pressure

Blood pressure (BP) is the force exerted by circulating blood on the walls of blood vessels and is influenced primarily by two factors, CO and PR. CO is the volume of blood the heart pumps per minute, and PR is the resistance to blood flow in the small arteries and arterioles. As shown in Figure 2, blood is pumped into the aorta during heart systole, and CO is decided at this stage. Then blood flows through the aorta, and the elastic arteries here give the blood circulation process capacitor-like properties, which is called arterial compliance. Large arterial compliance is mainly affected by age and will not change much in the short term. As blood moves into smaller arterioles and capillaries, it encounters resistance, which is denoted R. The Windkessel model [71] shows the relationship between BP and main factors.

$$BP \propto CO \times PR \quad (1)$$

It can be used to explain the variation of BP. For example, when a human feels cold, vasoconstriction will lead to an increase in PR and elevate BP.

2.2 Arterial Applanation Tonometry

Arterial applanation tonometry is a clinical method to record and analyze arterial pulse waveform[59]. As illustrated in Figure 3, the tonometry only needs to deploy a pressure sensor above the arterial and exert a proper external force P_e . Tonometry models the arterial as a cylindrical thin wall tube. According to Laplace's law of cylindrical tube, the arterial tension (T) is

$$T = \frac{P_t \times r}{\mu}, \quad (2)$$

where P_t is the transmural pressure, r is radius of wall curvature and μ is the arterial wall thickness. Therefore, when flattening the arterial wall with a pressure transducer, r approximates to infinity and P_t is close to zero. In this situation, the internal pressure P_i is equal to P_e . Then the output of the pressure sensor can exactly reveal the P_i of arterial, that is blood pressure. Although tonometry has the advantages of being non-invasive and easy to use, it should meet three strict requirements [44]. First, the artery measured should lie close to the skin and be anchored to a firm structure like a bone, such as a radius. Second, the sensor is consistently smaller than the area of the arterial wall being flattened. Hence, it needs a trained clinician to find the exact position due to the thin radial artery. At last, during the measurement, the position of the sensor should be fixed and ensure no physical movement. So, tonometry is not suitable to monitor blood pressure in the long term. In this paper, we leverage the tonometry to track BP for 30 seconds and the details are described in Section 5.2.1.

2.3 Hydrostatic Pressure in Artery

Hydrostatic pressure in the artery is the pressure that blood exerts at any given moment due to the gravitational force in a state of equilibrium. In general, the BP we measure is usually the pressure at the brachial artery, as its location is at the same height as the heart. Actually, the BP in different positions of the artery is different due to the influence of gravity [55]. Take the BP at the radial artery as an example, the local blood pressure here is denoted as P_{h0} when we put the wrist on the table. Then we raise our hand to the height of h_1 , and now the local BP is P_{h1} . Assume the central BP remains unchanged, the change in BP can be calculated as

$$P_{h1} - P_{h0} = \rho_{blood}g(h_1 - h_0), \quad (3)$$

where ρ_{blood} is the blood density and g is gravitational constant. Since the blood density is close to the water and varies little, a change in height of 1 cm will result in a local blood pressure change of 0.736 mmHg.

2.4 Arterial Blood Pressure Waveform and Pulse Pressure Waveform

The arterial blood pressure waveform (ABPW) is produced by cardiac ejection and complex reflection in pulsation propagation from the aorta to other arteries [49, 51, 57]. When the heart contracts, blood is ejected rapidly from the ventriculus sinister to the aorta and creates the systolic peak of the pulse waveform. As the pulse travels along the aorta to the peripheral arteries, the different vessel wall characteristics and bifurcation points will change the arterial resistance, resulting in the reflection and re-reflection. These reflections have an important influence on the ABPW, such as the form of the tidal peak due to the superimposition of forward wave and backward reflection waves during the systolic period. Hence, ABPW carries abundant hemodynamic information [8, 48, 67]. For instance, the maximum slope of the ascending and descending waveform reflects myocardial contractility and systemic vascular resistance, respectively. These two factors are highly related to CO and PR. The cumulative area under the ABPW curve indicates CO [48].

ABPW collected by invasive methods describes the pressure changes of blood flow inside the vessel, while PPW in tonometer is the pressure waveform measured from the outside of the arterial wall. Hence, according to the equal internal and external pressure mentioned in Section 2.2, PPW in tonometer can be treated as a substitute for ABPW. In practical terms, the high-frequency components of PPW may be lost due to skin and soft tissue damping. Considering this situation, we collect PPWs at the radial artery under a thin layer of skin to minimize distortion. Many previous studies have shown that the PPWs collected by tonometry are largely superimposable over the ABPW records measured by the invasive intra-arterial catheter [13, 59, 60]. The results in [60] showed that the waveforms collected by these two methods have the same spectral moduli from harmonics 1 to 6. Since ABPW is difficult to obtain, we infer BP with PPW in this paper.

3 PRACTICALBP DESIGN RATIONALE

In this section, we would like to elaborate the necessity of calibration for accurate cuffless BP measurement devices and the rationality of the personalized dataset generated by arterial applanation tonometry and hydrostatic pressure. At last, we will further explain the challenges we encounter.

The Necessity of Calibration. As discussed in Section 2.1, blood pressure is mainly influenced by CO and PR. And CO and PR vary greatly between individuals. The factors such as heart size as well as the force and the duration of heart contraction determine the value of CO [42]. For instance, the male usually has a larger CO than the female due to the larger heart size. PR is decided by vessel radius and length and elasticity of vessel walls and so on [20, 40]. The PR of the group that is 190 cm tall is about twice that of the group with a height of 160 cm [15]. Since BP and ABPW are jointly affected by so many individual physiological parameters, people with huge differences in BP may have similar ABPW, while people with similar BP may also have huge differences in ABPW [74]. Therefore, achieving an accurate cuffless BP measurement system by building a general model that does

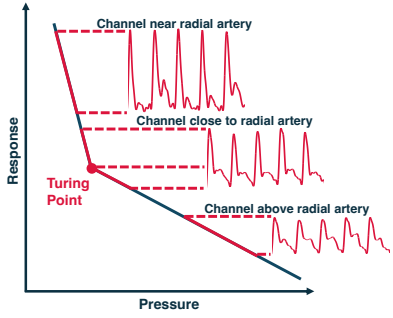


Fig. 4. The flexible pressure-sensitive sensor does not have a wide linear detection range.

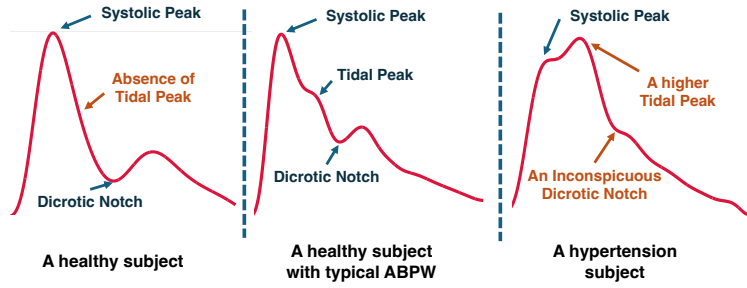


Fig. 5. Differences in ABPW between individuals.

not include the user's personal physiological information is an almost impossible task. Liu et al. [39] collected a large dataset with various user distributions. They found that even with such a large dataset, a calibration-free general model only achieved a high mean absolute error and standard deviation of 9.67 and 13.04 mmHg for SBP. The introduction of some static individual information such as height, gender, and weight can improve the BP measurement. However for dynamic information that is highly correlated with BP, such as the medication status of hypertensive, the performance of this method will degrade greatly. Therefore, PracticalBP directly embeds the user's CO and PR information into their personalized model.

The Rationality of Our Pseudo BP Labels. In order to improve the accuracy and generalization of the model, the personalized dataset must contain the user's CO and PR-related information. To this end, in our personalized dataset, arterial applanation tonometry is used to provide the user's CO-related information and arterial hydrostatic pressure carries a PR-related message. To be specific, the user will be asked to keep still at the stage of tonometry-based pseudo-label generation. The variation of chest pressure due to breathing will affect the blood volume of venous return, thereby changing the CO. It can be observed that the upper and lower envelope of ABPW changes synchronously with breathing [69]. Therefore, the user's CO-related information can be captured at this stage. For PR, as shown in Figure 2, the PR is modeled as a resistive component in the Windkessel model. The change of local hydrostatic pressure caused by arm tilting can be equivalent to adding a new series resistor and adjusting the total peripheral resistance. The arm tilting-based hydrostatic pressure is also applied in many studies [10, 38, 63, 64] to explore some important index for cardiovascular events. For instance, [38] leveraged it to assess arterial stiffness of subjects in different age groups and validated its correlation with pulse transit time (PTT). [63] studied the influence of changes in hydrostatic pressure on peripheral PPG waveforms and proved that the compliance state of a subject's vasculature can be extracted from it.

Challenge Understanding. First, generating accurate pseudo BP labels is challenging in practice. For tonometry, aiming to free from the participation of clinicians, we deploy a pressure sensor array that ensures PPW can be collected successfully. Next, we should find which channel is above the radial artery. This is essential to generate pseudo labels correctly. Existing work [52] customizes a multi-channel stress MEMS sensor and selects the channel with the highest amplitude of pulse waves based on the truth that the force response above the artery is the strongest. However, due to the need for external pressure, a hard MEMS sensor worn near the radial artery will cause discomfort. Therefore, we use a flexible film pressure sensor array to collect PPW as an alternative. But the film sensors are suffered from multiple linear regions and narrow linear detection range [3, 37, 41]. The different back pressure on sensors due to the unevenness of the skin interface will cause them to operate in

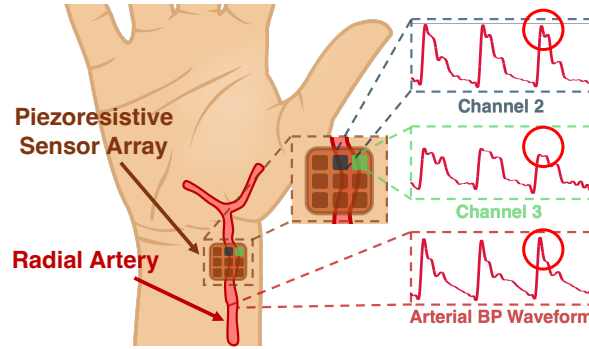


Fig. 6. The influence of sensor position on the collected pulse pressure waves.

different linear regions. Hence, the sensor near the radial artery may collect pulse waves with a larger amplitude than that of the sensor placed above the radial artery, as shown in Figure 4. That will make the existing solution fail and result in a huge deviation of tonometry-based pseudo labels. For hydrostatic pressure measurement, the IMU-based methods suffer from the cumulative error caused by drift, which will lead to unacceptable errors. For instance, an estimation error of 10 cm will bring an error of 7.4 mmHg in the pseudo label.

Second, it's necessary to understand the differences in PPW across the subjects. ABPW varies between individuals due to their specific Hemodynamic parameters. For example, Figure 5 shows three ABPW examples from three subjects. The middle one is a typical waveform that has a clear systolic peak, tidal peak, and dicrotic notch that is highly related to BP (as discussed in Section 2.4). And the ABPW of some healthy subjects may lack the tidal peak. Moreover, some individuals with cardiovascular disease (e.g., hypertension) have an ABPW with a higher tidal peak than a systolic peak and an inconspicuous dicrotic notch. That means the BP-related features of ABPW are different among individuals. Hence, the model should be aware of the different importance of these features in different subjects.

At last, although a sensor array is used, the discrete sensor distribution makes it possible for an artery to lie between adjacent sensors. Hence, the position-susceptible PPW will be distorted. As shown in Figure 6, channel 2 is very close to the radial artery, but not above it. Although its waveforms are very similar to the radial BP waveform, the relationship between the systolic peak and the tidal peak is distorted. In particular, when the user re-wears the device, the pattern of distortion caused by the change in position may be different each time. That will lead to unstable BP estimations due to noisy raw signals.

4 SYSTEM OVERVIEW

The overall procedure of PracticalBP is shown in Figure 7. The whole system has three stages that are calibration, personalization model training, and BP inference. When a new user comes, calibration will be conducted to build a personalized dataset. During the calibration, the user needs to wear our customized wristband which collects PPW and PPG. Then the user needs to perform the following operations in sequence, first, take a BP measurement with a cuff-based device, then place the hand wearing our device flat on the table and keep it still, and finally tilt the forearm to complete the arterial tonometry and hydrostatic pressure measurement. This process will last within 4 minutes. Next, the wristband will preprocess the collected PPW, PPG, and IMU signals, and generate the corresponding pseudo BP labels to form the personal dataset. After receiving the dataset sent from the wristband through a Wi-Fi module, the server will finetune the pretrained model and obtain a user-specific model. Finally,

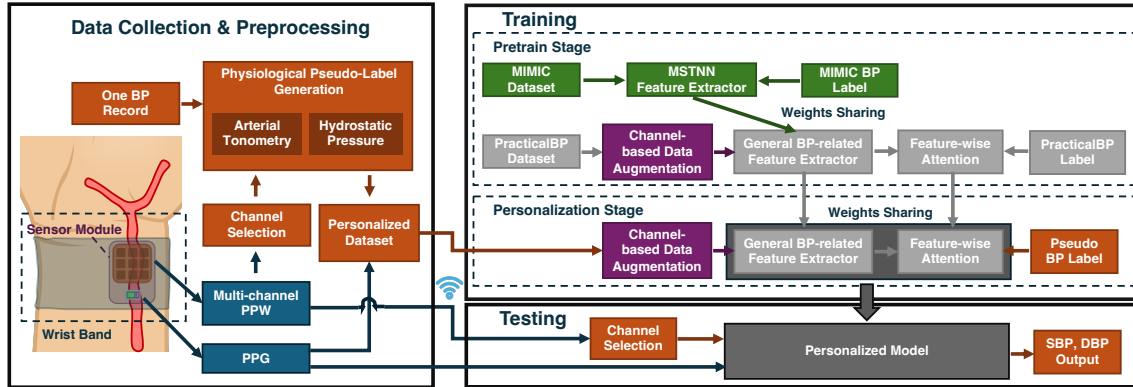


Fig. 7. System overview of PracticalBP.

when the user needs to measure BP, the device will collect and upload the PPW and PPG signals, and calculate the BP with the personalized model.

To introduce individual hemodynamic information accurately, we design a physiology-based pseudo-label generation mechanism with low effort. To be specific, a waveform-based channel selection algorithm and a discrete height measurement method are proposed for tonometry-based and hydrostatic pressure-based pseudo labels, respectively. For the sake of extracting effective features specific to the individual, we design two modules, a general BP-related feature extractor and a feature-wise attention mechanism. The first module will extract data-driven features and manual features with a large-scale dataset. And the feature-wise attention is designed to weigh the above features according to the user's characteristics. To improve the robustness of our system, a channel-based data augmentation (ChannelDA) is proposed to eliminate the impact of PPW distortion caused by the sensor position changes during multiple wears. This mechanism fully utilizes the multiple channels of PPW and augments the diversity of the training dataset effectively.

5 SYSTEM DESIGN

5.1 Sensor Module Design

In order to collect information about the force response on the vessel wall and changes in blood volume, we design a wristband with a sensor module containing a flexible piezoresistive sensor array and a PPG sensor. As shown in Figure 7, the sensor module is placed on the radial artery close to the radius. To meet the requirements of tonometry, each sensor in the array is customized to a dimension comparable to the diameter of the radial artery. For the PPG sensor, since the light emitted by it is semi-omnidirectional and reflects from many directions, only one PPG sensor is used and its placement requirements are relatively loose. The piezoresistive sensor array and PPG sensor are placed along the radial artery. The advantage of this placement is that blood flows first through the PPG sensor and then reaches the pressure sensor array, providing PTT information that is strongly correlated with BP. As discussed in Section 3, to collect stable signals, applying an external force on the sensor module is necessary. Therefore, we add a small airbag. Note that different from the oscillometric device that uses a cuff, our airbag is only used to provide back pressure on sensors. Our airbag only needs to exert a pressure of 60 mmHg, while the cuff-based devices require 150 mmHg or even higher pressure to occlude the artery. Hence, our wristband will not cause discomfort.

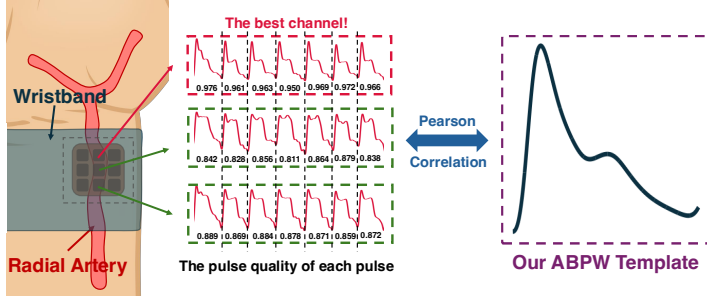


Fig. 8. Our ABPW template and an example of pulse quality calculation.

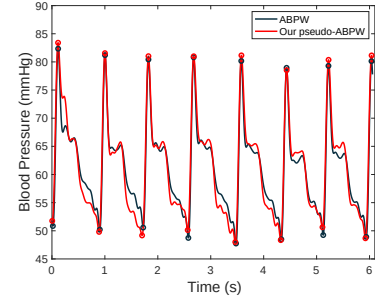


Fig. 9. The pseudo ABPW generated by tonometry.

5.2 Physiology-based Pseudo-Label Generation

In this section, we will explain how to generate accurate pseudo labels. The physiology-based pseudo labels include two parts. One part is produced by arterial applanation tonometry which contains the user's CO information and the other part is based on hydrostatic pressure in the artery which carries PR information.

5.2.1 Pseudo Labels Generation with Tonometry. As established in Section 2.2, tonometric measurements necessitate a compact sensor capable of effectively flattening the arterial wall. Therefore, the primary objective of this study is to identify an optimally positioned sensor within the array that satisfies these dimensional and functional requirements at the radial artery location. Since the signals from tonometry are usually used as an alternative to ABPW [47], the PPW collected by such a sensor should be consistent with the radial BP waveform. In light of this, we can determine the best channel by selecting the channel which signals have the highest similarity with the ABPW. However, it is not practical to obtain real-time ABPW for each user. To handle this, we observe that although the ABPWs vary from person to person, they all have a generally consistent pattern. Thus we can generate an ABPW template with the open dataset and our dataset. MIMIC-II [33] we used here is an open-access database that contains invasive ABPWs from 26870 adults. To be specific, we first segment pulses from the ABPW records in the MIMIC-II database as well as our dataset, and then regulate each pulse to a fixed length in order to eliminate the effect of heart rates. The ABPW template is the average of normalized pulses which is shown in Figure 8. Next, we can calculate the Pearson correlation coefficient between each pulse and the ABPW template to evaluate their similarity. The pulse quality of the i -th channel PQ_i is defined as

$$PQ_i = \frac{\sum_{j=1}^{\#Pulses} corr(Template, Pulse_{(i,j)})}{\#Pulses}, \quad (4)$$

where $Pulse_{(i,j)}$ is the j -th length-regulated pulse in the i -th channel and $corr(\cdot)$ is the Pearson correlation coefficient. For practical consideration, the signals of a piezoresistive sensor may fluctuate due to an unstable interface. We further introduce the standard deviation of systolic peak and onset value to penalize this situation. The channel quality of i -th channel CQ_i is calculated as follows,

$$CQ_i = \frac{PQ_i}{\sqrt{std(Systolic\ Peaks_i) \times std(Onsets_i)}}, \quad (5)$$

where $Systolic\ Peaks_i$ and $Onsets_i$ is the sequence of detected systolic peaks and onsets with zscore normalization of i -th channel and $std(\cdot)$ is the standard deviation. And the best channel is the channel with the highest CQ .

Pseudo Labels. After channel selection, we can get the pseudo labels with tonometry. The PPW of the best channel can be mapped to the ABPW through a linear transformation. To determine the linear mapping, it needs

Table 1. Comparison of amplitude-based and waveform-based channel selection.

Channel Selection Strategy	Correlation	MAE (mmHg)		ME (mmHg)		STD (mmHg)	
		SBP	DBP	SBP	DBP	SBP	DBP
Amplitude-based	0.85	6.35	4.21	1.21	0.24	6.64	4.21
Waveform-based	0.88	3.42	3.51	-0.06	0.14	3.73	3.90

at least one BP record $\{SBP_{cali}, DBP_{cali}\}$. Then, the slope k and bias c of the linear mapping can be expressed as

$$k = \frac{SBP_{cali} - DBP_{cali}}{SP_{best} - O_{best}} \text{ and } c = DBP_{cali} - k \times O_{best}, \quad (6)$$

where SP_{best} and O_{best} are the average of the first five detected systolic peaks and onsets in the best channel, respectively. Finally, the pseudo SBP SBP_{pseudo} and DBP DBP_{pseudo} is generated as followed,

$$\{SBP_{pseudo}, DBP_{pseudo}\} = k \times \{Systolic\ Peaks_{best}, Onset_{best}\} + c. \quad (7)$$

As shown in Figure 9, the pseudo ABPW generated with the best channel is in red and the ground truth ABPW is in black, and they have a high similarity of 0.978. The marked SBP and DBP of the two waveforms are very close. Table 1 shows the results of our pseudo label generation in our dataset. Compared with channel selection based on the pulse amplitude, our waveform-based strategy achieves a higher correlation with ground truth ABPW and a lower error of pseudo labels in both SBP and DBP.

5.2.2 Pseudo Labels Generation based on Hydrostatic Pressure. The key point to generating hydrostatic pressure-based pseudo labels is to measure the change in relative height. For this purpose, we deploy a six-axis IMU on the customized wristband. The attitude of the IMU can be solved by Mahony's algorithm and further obtain the rotation matrix. With the rotation matrix, the acceleration is converted from the sensor coordinate system to the ground coordinate system. Finally, the acceleration is integrated twice to obtain the displacement, and the displacement in the direction of gravity is the height. However, the relative height measurement based on IMU suffers from the cumulative error caused by drift. This will lead to an unacceptable error on generated pseudo labels. To tackle this problem, we notice that continuous location tracking is not necessary in our scenario. During this phase, the user will first place the forearm on the table, and then change the height of the wristband by tilting the forearm. The user will be asked to adjust the wristband to the height of Δh , $2\Delta h$, 0 , $-\Delta h$ and $-2\Delta h$ relative to the table and keep still for 30 seconds at each height. Therefore, in the whole procedure, the dynamic height change only happens 5 times and each duration is very short (about 1 second). The error comes from only two actions because when the relative height returns to 0, it is equivalent to a reset. Given this, we design a movement detection mechanism to eliminate drifts during the static state. Specifically, a threshold is set to segment the IMU data during the forearm tilting.

Pseudo Labels. The BP will not vary dramatically within a few minutes at rest. After 5 minutes rest, the average DBP and SBP variation is about 0.6 mmHg in 5 minutes [56]. Hence, we can still use the BP record in the beginning of calibration because our whole calibration procedure can be finished in three and a half minutes. According to Section 2.3, the pseudo label here is

$$\{SBP_{pseudo}, DBP_{pseudo}\} = \{SBP_{cali}, DBP_{cali}\} - 0.736 \times \Delta_{height}, \quad (8)$$

where Δ_{height} is in centimeter.

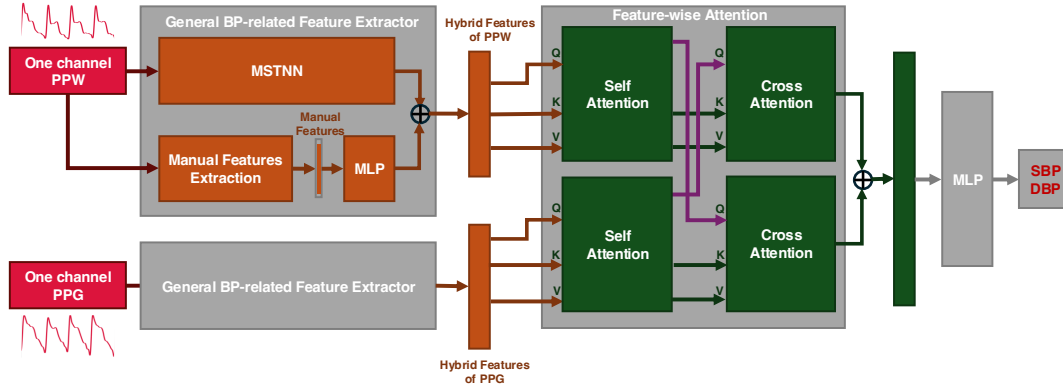


Fig. 10. The architecture of effective feature extraction.

5.3 Effective Feature Extraction

To extract features effectively, we propose two modules, general BP-related feature extractor and feature-wise attention. The first component is used to extract general features from a large dataset, and the second one can weigh these features and find the dependencies between them.

5.3.1 General BP-related Feature Extractor. To obtain the BP-related features adequately, we combine the data-driven features and manual BP-related features, as shown in Figure 10. First, for each modality, we leverage the spectro-temporal feature extractor in MSTNN [16]. This feature extractor is composed of four ameliorated convolutional blocks and a recurrent layer, which has fewer parameters and higher generalization capability than that in [65] and [34]. To improve the generalization of extracted features, we intend to leverage the knowledge of the open-source datasets. However, to the best of our knowledge, many datasets contain PPG, but no large-scale dataset contains PPW. Although many other cardiac activity-induced signals, such as ECG [16] and Seismocardiogram (SCG) [21] are highly correlated with BP, ECG is a bioelectric signal, and SCG is a mechanical vibration signal caused by heartbeat, which are very different from PPW in morphology and characteristics. Hence, they cannot be directly used for pre-training. To tackle this, recall that we discussed in Section 2.4, ABPW contains BP-related features consistent with PPW, thereby it can be used in pre-training. As illustrated in Figure 7, in the pretrain stage, we first train the feature extractor with normalized ABPW and PPG in the well-known MIMIC-II [33] dataset and then continue to train it with PPW and PPG in our dataset in order to adapt the actual pattern of the signals. The normalization of ABPW is to eliminate BP values that are not carried in PPW.

Then, many previous researches have verified that many manually extracted features have a high relation with BP, and this motivates us to combine the human knowledge that supplements the data-driven features. To be specific, we select 27 features mentioned in [14, 16] for each pulse in PPW and PPG. The detailed definition of these features is described in Appendix A. To improve the stability and eliminate the effect of heart rate, these features are extracted from pulses which are normalized in terms of amplitude and time dimension. Then we average the features of multiple pulses in each sample, thus the dimension of the manual features of each sample is 27×1 . Next, an MLP is used to expand the dimension to align with the dimension of data-driven features. At last, we concatenate the data-driven features and manually extract features to generate hybrid features.

5.3.2 Feature-wise Attention. Different from FewShotBP [16] that use an MLP directly, we use the attention mechanism to let the model further capture the dependencies between features.

To this end, we integrate a self-attention module and a cross-attention module. And the former is to connect the features within the single modality, the latter is to fuse the features of PPW and PPG. The feature-wise attention module has three inputs, X_q , X_k , and X_v that all have a shape of $N_f \times d_f$, where N_f is the number of features and d_f is the dimension of each feature. Then the output of the module F_{out} can be calculated as followed,

$$Q = X_q W_q, K = X_k W_k, V = X_v W_v, \quad (9)$$

$$S = \text{softmax}\left(\frac{QK^T}{\sqrt{d_f}}\right) \text{ and } F_{out} = SV, \quad (10)$$

where Q , K , and V are the query, key, and value of the attention module, and each of them has a learnable matrix with a shape of $d_f \times d_f$, W_q , W_k and W_v , respectively. S is the attention score and contains the inner product of Q and K , then the dependencies between features in Q and K can be learned. At last, F_{out} is the weighted V . As shown in Figure 10, in our feature-wise attention mechanism, the feature matrix of PPW and PPG are first fed into a self-attention module, respectively, where X_q , X_k and X_v are equal. Then, a cross-attention module is used to learn the relationship between PPG and PPW features. Specifically, the PPW features fused with PPG features are obtained by the attention module where X_q is the weighted PPG features matrix, X_k and X_v are the weighted PPW feature matrix, and vice versa. This procedure is repeated twice. Finally, the weighted PPW and PPG feature matrices are concatenated together and the estimated BP is inferred by an MLP.

5.3.3 User-specific Feature Extraction in Personalization Stage. At the personalization stage, the individual dataset generated in Section 5.2 is used to finetune the pretrained model. While the morphology of PPW and PPG demonstrates consistent anatomical patterns across individuals, significant interindividual variations in vascular characteristics necessitate model customization to accommodate user-specific physiological parameters. Therefore, during finetuning, we freeze the parameters of the first three convolutional blocks in MSTNN. The last block has a narrower receptive field which corresponds to the subtle features, thus it should be learnable to adapt user-specific pattern. And our feature-wise attention mechanism at this stage can readjust the weight of features. It is crucial because some features may be absent in some pulse waves. For example, some healthy subjects have a missing tidal peak.

5.4 Channel-based Data Augmentation

ChannelDA is designed to make our system immune to distorted PPW caused by inaccurate sensor position and improve practicality. As discussed in Section 3, the PPW is susceptible to the sensor position and then leads to unstable features. Although we have nine channels of PPW and can try to reconstruct the high-fidelity ABPW, the inconsistent sensitivity of the piezoresistive sensor in the array makes it difficult to achieve. Given this, we instead let the model focus on features that remain consistent across different sensor locations to improve the stability of the model's predictions. To be specific, we use the best channel as the basis and combine other channels to generate a large amount of virtual PPW signals at different positions, and feed these data to the model so that the informative features from these distorted signals can be learned. To this end, we can reconstruct the PPW at locations between array units by linearly combining their signals. Since we only input the best PPW channel in use, we only need to reconstruct the PPW signal around the best channel. As described in Figure 11, the virtual PPW is a weighted sum of the best channel and a random channel. The random weights of two channels follow two beta distributions which are shown in Figure 12. The beta distribution is determined by two parameters, α and β . The weights in the beta distribution (α and β) for the best channel are concentrated in the area close to 1 which makes the generated PPW around the best channel. The weights of the random channel are under the distribution with the same α and β of 0.5. Its weights are mainly distributed in the area close to 0 or close to 1 which can improve the authenticity of the synthetic signals. During training, for each sample, the best channel of PPW and two virtual PPW will be fed into the model mentioned in Section 5.3 in turn. For

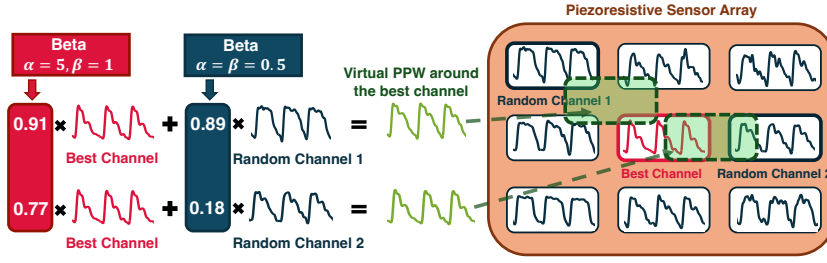


Fig. 11. The generation of virtual PPW around the best channel in ChannelDA.

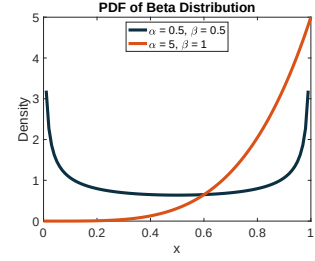


Fig. 12. Beta distribution with different parameters.

every epoch, the PPW channels used for synthesis are randomly selected again, and the synthesis weights are randomly generated again. The loss function is defined as follows.

Loss Function. The estimated error in loss function is calculated by smoothL1loss $\mathcal{L}_{Smooth}$, which can prevent exploding gradients [58]. The hyperparameter β is set to 5. With our ChannelDA, the loss function is defined by

$$\mathcal{L}_{ChannelDA} = \frac{\sum_{c \in C} \mathcal{L}_{Smooth}(y_c, \bar{y})}{\#Channel}, \quad (11)$$

where C stands for the set of input channels, y_c is the prediction of the c -th channel, and $\bar{y}$ is the actual BP which shares with all channels in a sample. Next, to further improve the stability of feature extraction, we add a neighbor prediction penalty $\mathcal{L}_{neighbor}$. It's based on the truth that BP will not vary a lot in a few seconds. Hence, we divide a sample X into three overlapping small samples, x_{n-1}, x_n and x_{n+1} . Then, for channel c , $\mathcal{L}_{neighbor}$ can be written as

$$\mathcal{L}_{neighbor}(c) = \frac{1}{2}(\mathcal{L}_{diff}(y_{n-1,c}, y_{n,c}) + \mathcal{L}_{diff}(y_{n,c}, y_{n+1,c})) \quad (12)$$

with

$$\mathcal{L}_{diff} = \begin{cases} 0, & \text{if } |a - b| < t, \\ |a - b| - t, & \text{otherwise,} \end{cases} \quad (13)$$

where t is the threshold and set to 4. When the difference between the predictions of adjacent samples is larger than t , the penalty will increase linearly. At last, our overall loss function is defined as

$$\mathcal{L}_{overall} = \frac{1}{3} \sum_{x \in X} \mathcal{L}_{ChannelDA}(x) + \frac{1}{3} \sum_{c \in C} \mathcal{L}_{neighbor}(c). \quad (14)$$

6 IMPLEMENTATION

6.1 Customized Sensing Wristband

We implement PracticalBP with a customized wristband that comprises a sensor module, a main circuit, and a nylon strap, as shown in Figure 13. The sensor module contains a custom 3×3 piezoresistive sensor array (SuZhou LeanStar Electronic Technology Co., Ltd) and a commercial MAX30101 PPG sensor to collect the user's PPW and PPG signals, respectively. The size of the film sensor array is $21 \text{ mm} \times 21 \text{ mm} \times 0.5 \text{ mm}$, in which each element has a $4 \text{ mm} \times 4 \text{ mm}$ dimension that is comparable with the diameter of the radial artery. Moreover, the ultra-thin thickness ensures excellent flexibility and wearing comfort. The PPG sensor can emit light in three bands: red, green, and IR. PracticalBP only uses the IR band, since we found in practice that only the IR band obtains stable blood volume change signals in all subjects due to its strongest ability to penetrate the skin. These two sensors are attached along the radial artery on a custom TPU airbag.



Fig. 13. The prototype of our wristband and experiment setting.

The key components of the main circuit involve an STM32F407ZGT6 MCU, an MZB3004T04 micro pump, an LHDA1231415H micro valve, an MPU-6050 IMU, an ESP32 WROOM Wi-Fi module and some driving circuits. The micro valve and pump are used to control the back pressure in the airbag. They are connected to the airbag with silicone tubes. The six-axis IMU can capture the motion information to measure the relative change of height during hydrostatic pressure-based pseudo BP labels generation. The MCU controls main components such as micro valve and pump and collects PPW, PPG, and IMU signals. The sensor module and main circuit are connected with FPC lines. The multi-channel PPW signal is obtained in a time-division multiplexing manner based on a simple resistor voltage divider and two ADG704BRMZ multiplexers. Each PPW channel, PPG, and IMU are sampled at a high rate of 1600 Hz, which ensures the high time resolution of manual features. All collected data are transmitted to the PC by the Wi-Fi module. At last, the sensor module and the main circuit are adhered to on a nylon strap to compose our wristband. The total cost to manufacture a wristband is around \$61, of which the sensor module and main circuit cost \$17 and \$43, respectively. And the cost will be further reduced for large-scale production. The power consumption of our wristband is around 200mW, and it can work continuously for one and a half hours with a small 85 mAh battery. Note that since our system only needs 5 seconds of data to get a measurement, around 2160 BP measurements can be obtained when fully charged.

6.2 Data Collection

We recruit 43 participants and record 73175 BP values to evaluate PracticalBP on the basis of research protocols approved by the institutional review board. The subjects include 25 healthy subjects and 18 patients diagnosed with hypertension. The detailed demographics of our dataset are described in Table 2 and the distribution of collected BP is illustrated in Figure 15. Among the patients, some of them are taking antihypertensive drugs to control their BP. The BP ground truth is collected by CNAP Monitor 100D [6] which is an FDA-approved continuous blood pressure measurement device. And the data is real-time read by the BIOPAC MP160 system and AcqKnowledge software [5]. Before the data collection, as shown in Figure 13, the subjects will first wear our wristband and CNAP finger cuff on their right hand and the cuff for CNAP calibration on their right arm. And then they are asked to sit calmly with their hands flat on the table for 5 minutes. Figure 14 shows the procedure of our data collection which has two stages, the calibration stage and the test stage. In the calibration stage, the

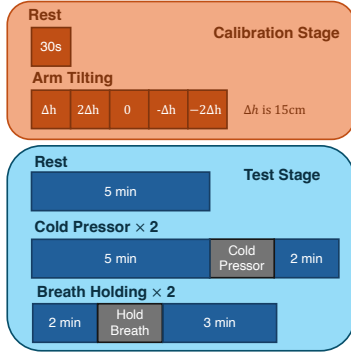


Fig. 14. The protocol of data collection.

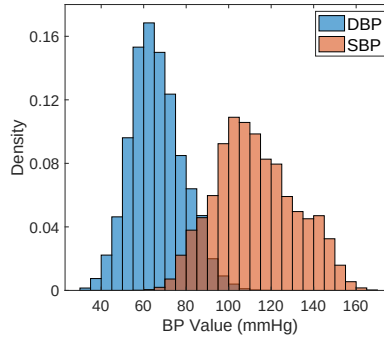


Fig. 15. The BP distribution of our dataset.

Table 2. Demographics of subjects.

Info	Stats
#Subjects	43
#Patients	18
Gender	27 M & 16 F
Age (years)	37.3 (17.5)
Weight (kg)	66.7 (11.2)
Height (cm)	168.8 (9.7)
BMI (kg/m ²)	23.4 (3.6)
SBP (mmHg)	113.1 (19.2)
DBP (mmHg)	67.2 (13.2)

Statistic representation: Mean (STD).

subject will first measure cuff-based BP once and then stay still for 30 seconds for tonometry-based pseudo-label generation. Next, for hydrostatic pressure-based pseudo labels, the subject will be asked to tilt his/her arm to the height of Δh , $2\Delta h$, 0 , $-\Delta h$ and $-2\Delta h$ relative to the table in turn. And the subject should keep the arm still for 30 seconds at each height. Therefore, the whole duration of this stage is within 4 minutes. During the test stage, a 5-minute rest group will be conducted, and then two BP intervention experiments that are cold pressor and breath holding will be conducted in two groups separately. The cold pressor test will stimulate the sympathetic nerves and cause vasoconstriction, ultimately leading to a transient increase in BP [31]. In this group, the subject will first rest for 5 minutes and then put the left hand in an ice water container for the longest time the subject can keep. At last, place the hand back on the table and stay calm for 2 minutes to wait for BP to return to normal. And breath holding will affect venous return and thus reduce stroke volume, causing arterial BP to rise briefly, and then fall, and finally return to normal after respiration is resumed [28]. In detail, the subjects are required to rest for 2 minutes first and begin to hold their breath until can no longer resist, and then resume breathing and wait for 3 minutes. Note that only healthy subjects will conduct the BP intervention experiments and hypertensive patients perform four 5-minute resting groups instead.

6.3 Learning Details

We implement our neural network models with Python 3.7 and PyTorch 1.10. All evaluation experiments are conducted on one NVIDIA GeForce RTX 4090 GPU. To match the sampling rate of data in the MIMIC-II dataset, the raw data is downsampled to 125 Hz. The inputs of PPW and PPG are 5-second segments. In the training stage, since our neighbor prediction penalty, the signals are first windowed into frames of 9 seconds with a 6-second overlap, and then each 9-second frame is windowed into three 5-second frames with a 3-second overlapping. In the testing stage, we use the 5-second sliding window with a 3-second overlap directly. For the output layer after the feature-wise attention, three fully connected layers, Linear(384,192), Linear(192,64), and Linear(64,2) are used. We set an initial learning rate to be $1e-3$ and $1e-5$ for pretrain stage and personalization stage, respectively. A MultiStepLR scheduling scheme with a gamma of 0.5 and an MSprop optimizer is used in both stages. We set the batch size to 256 and 128 and train the model across 60 and 100 epochs for pretrain stage and personalization stage, respectively. To assess our system effectively, all of our performance is evaluated under the leave-one-subject-out scheme.

Table 3. Overall performance of PracticalBP compared with baselines. (Metrics in mmHg). Our results are in **Bold**.

Metrics	AAMI		PracticalBP		FewShotBP I		FewShotBP II		BP ³	
	SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP
MAE	-	-	6.19	4.78	5.69	4.39	7.98	5.75	9.91	8.14
ME	≤ 5	≤ 5	-1.49	-0.83	-0.05	0.06	-1.30	-1.48	0.94	0.45
STD	≤ 8	≤ 8	7.89	6.23	7.35	5.85	10.15	7.56	12.29	10.07
PCC	-	-	0.91	0.86	0.92	0.88	0.85	0.80	0.73	0.56

7 EVALUATION

7.1 Evaluation Methodology

7.1.1 Baseline Selection. We select BP³ [74] and FewShotBP [16] as the baselines. Both these two works introduced personal information to achieve the cuffless BP measurement. And they are evaluated on our dataset. For BP³, we try our best to reproduce its physio-pulse attention mechanism and feed the best three PPW channels in our piezoresistive sensor array and subjects' physiological factors including gender, age, BMI, and history of hypertension as input. For FewShotBP, we implement it with its open-source code. And the input is the best channel of PPW and PPG signals. Ten actual BP values are used to train the personalization adapter. Two types of it will be implemented for a comprehensive comparison. FewShotBP I uses BP records in both rest and BP intervention experiments to train the personalization adapter, which has the same setting in the original paper. And FewShotBP II only uses BP labels in the rest experiments which stands for a setting with less effort. Note that the input length of our work and FewShotBP is 5 seconds and those of BP³ is 30 seconds which is consistent with the original paper.

7.1.2 Performance Metric. Four common metrics, mean absolute error (MAE), mean error (ME), standard deviation error (STD), and Pearson's correlation (PCC) are used to evaluate our BP measurement. They are defined as follows, $MAE = \frac{\sum_n^N |\bar{y}_n - y_n|}{N}$, $ME = \frac{\sum_n^N (\bar{y}_n - y_n)}{N}$, $STD = \sqrt{\frac{\sum_n^N (\bar{y}_n - y_n)^2}{N}}$ and $PCC = \frac{\sum_n^N (y_n - \mu_y)(\bar{y}_n - \mu_{\bar{y}})}{\sqrt{\sum_n^N (y_n - \mu_y)^2 \sum_n^N (\bar{y}_n - \mu_{\bar{y}})^2}}$, where N is the number of samples, y_n and $\bar{y}_n$ are the BP predict of n -th sample and its ground truth and μ_y and $\mu_{\bar{y}}$ are the mean values of BP predicts and ground truth BP values.

7.2 Overall Performance

We will first demonstrate PracticalBP's overall performance by comparing it with the baselines. The results are shown in Table 3. Our system achieves a ME and STD of -0.83 and 6.23 mmHg for DBP, -1.49, and 7.89 mmHg for SBP, which both meet the standard requirements of AAMI. Compared with FewShotBP I, PracticalBP achieves a similar performance. However, the time cost of calibration in our system is less than one-fifth of that in FewShotBP and our system does not need specific labels for BP intervention. That proves the effectiveness of our physiology-based pseudo-labels. In addition, we further explore the performance of FewShotBP in the calibration setting using only resting BP labels. Due to the lack of BP variation labels, FewShotBP II cannot fully capture the user-specific cardiovascular information and its STDs exceed the AAMI requirements. Moreover, in contrast with BP³, our system has a significant improvement in all metrics. The main reason is that simple personal information introduced by BP³ cannot fully consider the complex factors that affect BP such as medicine use. We find that the hypertensive patients we recruited often take antihypertensive drugs according to doctor's orders, which makes the their SBP records not all exceed 140 mmHg. Due to the lack of this key information, BP³ overestimates the BP of these confirmed hypertensive patients, while PracticalBP directly uses the underlying

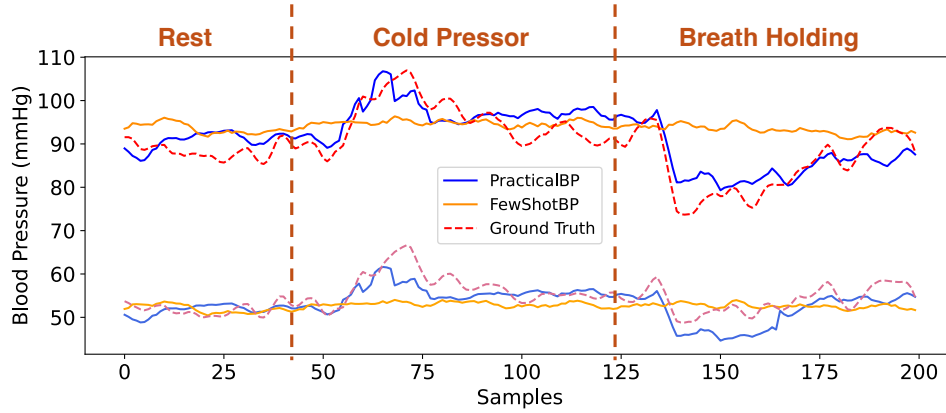


Fig. 16. An example of continuous BP measurement. BP³ is not shown in this figure due to different input length.

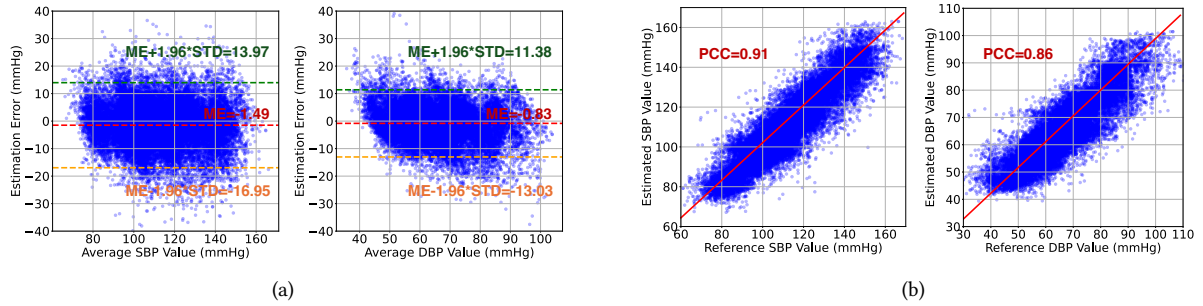


Fig. 17. (a). Bland-Altman diagram of PracticalBP; (b). Correlation diagram of PracticalBP.

hemodynamic information to model and achieve better performance. In terms of continuous measurement in time series, Figure 16 displays an example that contains the resting and BP intervention experiments. We can see that PracticalBP can track the variations in both SBP and DBP well while FewShotBP I is not sensitive to the BP change despite estimating accurately in resting. That demonstrates our design effectiveness in user-specific feature extraction.

To comprehensively examine our performance, we also plot the Bland-Altman diagram and correlation diagram of PracticalBP. In Figure 17 (a), the x-axis is the average estimated BP values and ground truth values, while the y-axis represents the estimation error. The dashed lines in red and green are the limits of agreement, which indicates that over 95% of points fall within these limits. Figure 17 (b) illustrates the correlation between our BP estimation with reference BP values and PracticalBP has a high PCC of 0.91 and 0.86 for SBP and DBP, respectively. These two diagrams demonstrate that our system can achieve a consistent estimation performance across the wide BP range. What's more, we present our system's performance on each subject as well. As described in Figure 18, we can observe that the results of all subjects achieve relatively consistent performance. Notably, there are 5 subjects whose MAE of SBP is larger than 8 mmHg. We find that these five subjects are all from the hypertensive group and the average age reaches up to 61.6. Hence, this phenomenon may be a result of arterial stiffening due to aging.

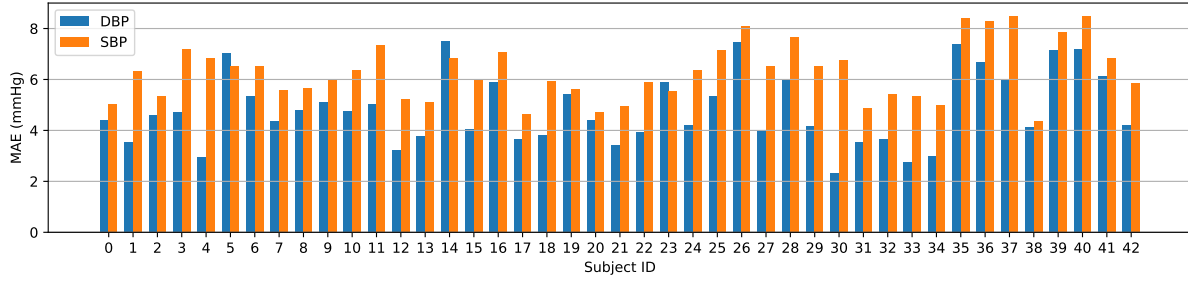


Fig. 18. Subject-wise Performance. Subject 0-24 are in the healthy group and Subject 25-42 are hypertensive patients.

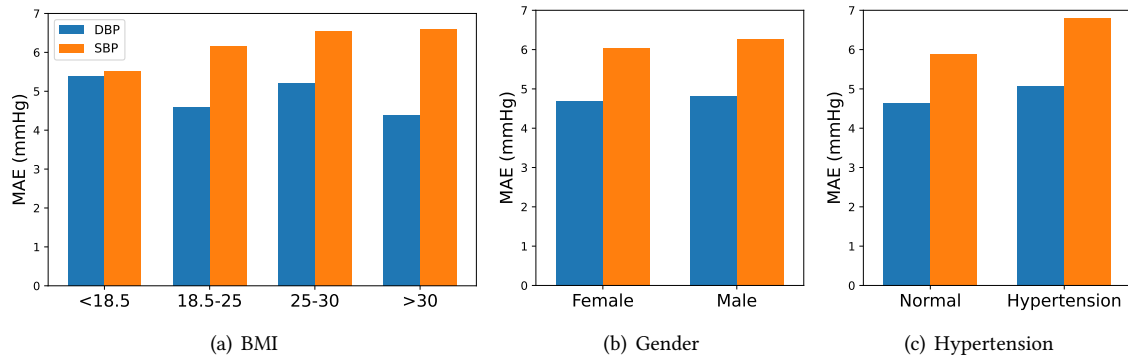


Fig. 19. Demographic Analysis. (a) Impact of BMI. (b) Impact of gender. (c) Impact of disease.

7.3 Demographic Analysis

Next, we will explain the impact of different demographic factors on PracticalBP. We will mainly discuss three factors, BMI, gender, and disease. The MAE of SBP and DBP are used to analyze here.

7.3.1 Body Mass Index (BMI). We first examine the performance across different BMI groups. As shown in Figure 19 (a), the subjects are divided into thinness (< 18.5), normal ($18.5 - 25$), overweight ($25 - 30$) as well as obesity (> 30). From the results, we can observe that the MAE of SBP increases slightly with increasing BMI. The main reason is that the pulse of overweight individuals is often difficult to capture since fat increases the distance between the radial artery and skin, which results in low-quality PPW and PPG. In contrast, the pulse of leaner individuals is more pronounced and easier to perceive. What's more, the overweight usually has a higher BP and a larger BP variation. But generally speaking, the MAE in the overweight and obesity group is still acceptable.

7.3.2 Gender. Gender is an important factor affecting BP. Statistically speaking, men tend to have higher blood pressure levels than women, especially in younger age groups. Moreover, compared with men, women have thinner radial arteries, which makes it more difficult for us to collect high-quality PPW. According to the results shown in Figure 19 (b), our system performance in female and male groups has no significant difference, demonstrating the robustness of PracticalBP to gender.

7.3.3 Hypertension. We further analyze the impact of hypertension on the performance of our system. We divide the subjects into two groups based on the hospital's confirmed records, normal and hypertensive. According to Figure 19 (c), both SBP and DBP MAE of the hypertensive are slightly higher than that of the normal. The

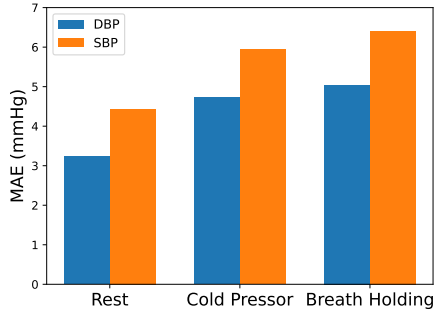


Fig. 20. Impact of each BP intervention experiment.

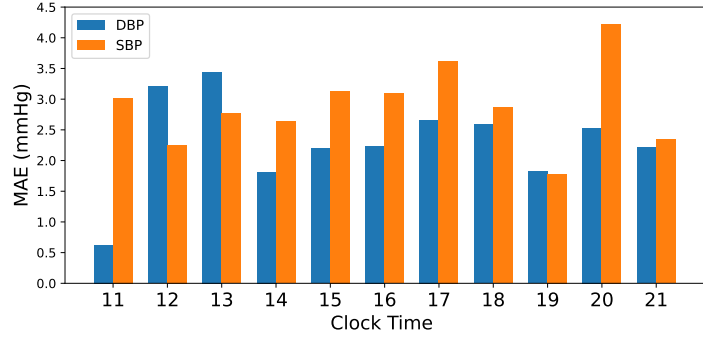


Fig. 21. Performance measured at hourly intervals in a day.

reason behind this may be the larger BP variation of hypertension. Specifically, Hypertensive patients often have decreased vascular elasticity which makes the blood vessels less responsive to variations in blood flow. In addition, many of them will take antihypertensive medications to regulate their BP. In our dataset, the average STD of ground truth values in the hypertension group are 6.98 and 5.68 mmHg for SBP and DBP, while those in the normal group are only 4.38 and 3.28 mmHg, respectively. For all this, PracticalBP still achieves a ME and STD in the hypertension group of -2.05 and 8.62 mmHg for SBP, -1.23, and 6.68 mmHg for DBP, which is satisfactory.

7.4 Robustness Study

In this section, we will further explore the robustness of our system. The performance of our system in each BP intervention experiment will be described. Then we deploy a seven-day experiment to verify the effectiveness of our personalized dataset in the time dimension. At last, we evaluate the effect of the relative height of wrist in BP measurement.

7.4.1 BP Intervention. As mentioned in Section 6.2, we conduct cold pressor and breath holding to adjust the BP of health subjects. And the former can elevate BP in a short time by changing PR and the latter will reduce BP temporarily by adjusting CO. The results are shown in Figure 20. Compared with the resting experiment, the estimated errors of both BP intervention groups increase, with the MAE of SBP and DBP in cold pressor adding about 1.5 mmHg and the MAE of SBP and DBP in breath holding increasing by 1.80 and 1.97 mmHg, respectively. Despite this, the MAE of SBP is still lower than 7 mmHg, which is acceptable. Moreover, from Figure 16, our system can track the change of BP correctly in both BP experiments. All in all, these results can demonstrate the robustness of PracticalBP in situations where BP changes dramatically.

7.4.2 Valid Duration of Personalized Model. To verify the validity time of the personalized model obtained through a single calibration, we conduct an hourly experiment and a one-week experiment on three subjects, including two normal subjects and one hypertensive patient.

Hourly Evaluation. In this experiment, the subjects are asked to calibrate and generate their personalized model at 10 am, and then measure BP every hour using our device and an oscillometric device from 11 am to 9 pm on the same day. Each measurement collects 30-second data. Since there is only one discrete BP ground truth, we use the average prediction value of our system within 30 seconds as the output. The results are shown in Figure 21 and we can see that the estimation error of SBP and DBP fluctuates below 4.5 mmHg, and there is no trend of the MAE increasing over time, demonstrating the effectiveness of our individual models in a day.

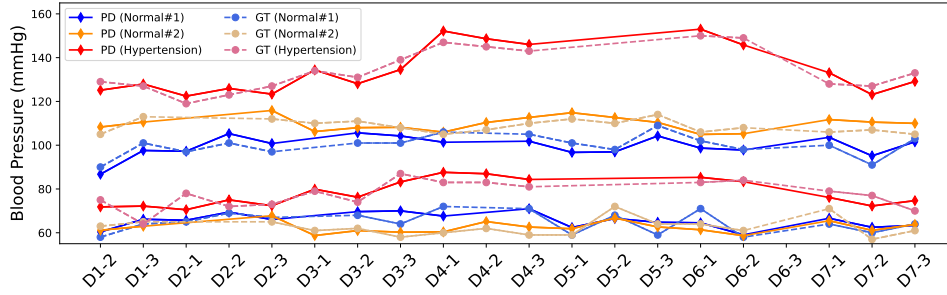


Fig. 22. The BP measurements of PracticalBP within a week. D2-1, D2-2, and D2-3 stand for the morning, noon, and evening of the second day, receptively, and so on. PD is the prediction of our system and GT is the ground truth.

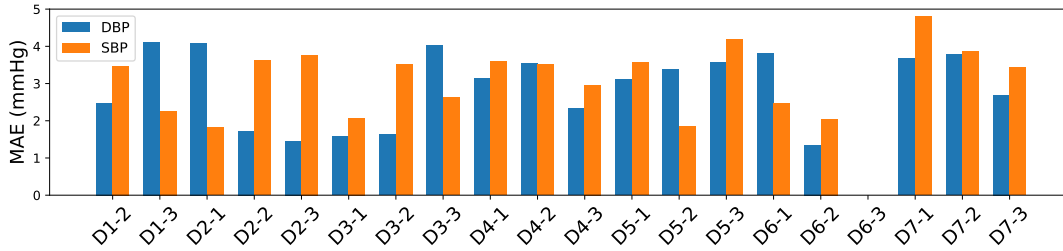


Fig. 23. One week performance.

One-week Experiment. In terms of the seven-day experiment, the subjects will execute calibration in the morning of the first day, and then measure BP once in the morning, noon, and evening in the next seven days, respectively. The BP measurements of three subjects are shown in Figure 22 and missing data caused by the absence of subjects due to some reasons will not be displayed in the figure (e.g., D6-3). From the results, we can observe that although the BP values of the hypertensive patient show a greater variation than those of normal subjects, our system can still track the trend well. The MAE of DBP and SBP are 3.14 and 2.51 mmHg for normal subject#1, 2.98 and 2.71 mmHg for normal subject#2, 3.34 and 3.54 mmHg for the hypertensive. The MAE of three subjects in a week is illustrated in Figure 23. We can see that the MAEs are less than 5 mmHg in a week and the error of our system does not increase significantly on the seventh day. This means that our system only requires users to spend four minutes to calibrate and can obtain accurate BP measurements for at least seven days.

7.4.3 Relative Height of Wrist in BP Measurement. The height difference between the wrist and the heart affects hydrostatic pressure at the radial artery. Considering users may measure BP while seated in chairs of varying heights after calibration, we conduct an additional experiment to investigate its impact on our system. Two subjects adjust their chair height so that the angle between the upper arm and the body is 45 degrees while their forearms rest flat on the table. After this calibration, the chair height is further adjusted to create bicep angles of 0 degrees and 90 degrees, representing two extreme cases. The experimental results indicate that the MAE of SBP for PracticalBP is 5.4 mmHg, 2.5 mmHg, and 5.8 mmHg at angles of 0 degrees, 45 degrees, and 90 degrees, respectively. This suggests that when the angle deviates from that during calibration, the system's MAE increases by approximately 3 mmHg which is acceptable. This increase can be attributed to the average forearm length of an adult, which is about 35 cm. Typically, when the arm is naturally hanging, the heart is situated at the

Table 4. Ablation Study. (Metrics in mmHg). The best results are in **Bold**. The first column shows the results when removing the main designed modules and the second column depicts the detailed ablation study. PSL stands for physiology-based pseudo labels and H-PSL is hydrostatic pressure-based pseudo labels. EFE is effective feature extractor. H-PSL ($2\Delta h$) is the hydrostatic pressure-based pseudo labels when arm titling is at height of $2\Delta h$, P-EFE represents the pretrained feature extractor with open dataset MIMIC-II and MF-EFE is the manually extracted features in effective feature extractor.

	PracticalBP		w/o PSL		w/o H-PSL		w/o EFE		w/o ChannelDA	
Metrics	SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP
MAE	6.19	4.78	13.80	8.57	8.41	5.93	8.34	5.81	6.97	5.26
ME	-1.49	-0.83	-2.85	-1.97	-2.67	-1.42	-2.51	-1.69	-0.15	0.07
STD	7.89	6.23	17.48	11.01	11.35	7.96	10.72	7.39	9.1	6.87
PCC	0.91	0.86	0.46	0.48	0.81	0.77	0.83	0.81	0.88	0.84
	w/o H-PSL ($2\Delta h$)		w/o H-PSL ($-2\Delta h$)		w/o H-PSL ($\pm 2\Delta h$)		w/o P-EFE		w/o MF-EFE	
Metrics	SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP	SBP	DBP
MAE	6.33	4.93	6.20	4.82	6.32	4.88	6.86	5.12	6.80	5.00
ME	-1.96	-1.29	-1.32	-0.50	-1.76	-0.95	-1.46	-0.93	-1.12	-0.69
STD	7.97	6.39	7.91	6.25	7.98	6.37	8.98	6.66	8.89	6.53
PCC	0.91	0.85	0.91	0.86	0.90	0.85	0.88	0.84	0.89	0.85

midpoint of the forearm's height. Consequently, a change in bicep angle results in a relative height variation of approximately 5 cm, corresponding to a hydrostatic pressure change of 3.68 mmHg. Although chair and table heights are generally not designed to maintain bicep angles of 0 or 90 degrees for comfort, we recommend that users be instructed to adjust their chair height before measurement to achieve a bicep angle close to 45 degrees, thereby minimizing this impact.

7.5 Ablation Study

We further conduct ablation experiments to understand the contributions of PracticalBP designs, specifically the physiology-based pseudo-label generation, effective feature extraction, and channel-based data augmentation. The results are shown in Table 4. The metrics including mean absolute error, mean error, the standard deviation of errors, and Pearson's correlation are used.

7.5.1 Benefit of Physiology-based Pseudo Label Generation. To validate the effectiveness of our physiology-based pseudo BP labels, we skip the model personalization stage and only use the pretrained model to estimate BP. As shown in the second column of the first row of Table 4, the MAE of SBP and DBP almost double. And STD of SBP and DBP increase by 9.59 and 4.78 mmHg, respectively, which exceeds the requirements of the AAMI standard. Furthermore, Pearson's correlations are less than 0.5, which means pretrained model fails to measure BP in a wide BP range. This phenomenon is supported by the previous conclusion in Section 3 that BP cannot correspond one-to-one to the features extracted from the input waveform due to the complexity of factors affecting BP.

We further conduct an experiment in which only tonometry-based pseudo labels are included. According to the third row, compared with the complete system, the MAE of BP measurement increase by about 30% and the STD of SBP and DBP increase by 3.46 and 1.73 mmHg, respectively. From this, we can conclude the following. (1) Even if only tonometry-based pseudo labels that lack BP variation are introduced, the accuracy of BP measurement will be greatly improved. This is because the personalized model carries the unique features of a specific user. (2) The dataset generated by arm tilting can correctly reflect the pattern of real BP variation and the PR information introduced by hydrostatic pressure-based pseudo labels is significant.

Table 5. Computational cost for deployment.

Stage	GPU Memory	Latency
Personalization	3832 MB	240 s
Inference	2126 MB	74 ms

Next, we evaluate the situation where some hydrostatic pressure-based pseudo labels are missing. In actual use, some users' arms may tremble during the arm tilting process due to weak muscle strength, resulting in low-quality data. We try to delete the pseudo labels obtained at a farther distance from the desktop, and the results are shown in the second row of Table 4. In contrast with the complete system, the MAE of BP measurement increase slightly and the STDs of SBP error are still less than 8 mmHg which meets the requirements of AAMI standard.

7.5.2 Benefit of Efficient Feature Extraction. To evaluate our effective feature extractor mechanism, we remove the manual features, open dataset part, and our feature-wise attention module. The features extracted by MSTNN are fed to an MLP directly. Based on the comparison shown in the first and fourth rows of the first row in Table 4, we can see that our effective feature extractor also contributes to the overall performance. Without it, the MAE of SBP and DBP increase 2.15 and 1.03 mmHg, respectively. The STD of SBP increases by 2.83 mmHg. The reason why the STD degrades a lot is that the lack of a large-scale dataset and empirical features leads to not generalized enough feature extraction. The differences in PPW and PPG waveforms between individuals lead to differences in feature patterns, which cause some unstable features to perturb BP prediction. Our attention mechanism can weigh the features based on user-specific signals, which benefits the BP measurement. A more detailed ablation study is shown in the second column of Table 4. We can see that whether removing pretraining based on the large-scale dataset or eliminating manually extracted features, the performance of the system has dropped significantly. The STD of their SBP error reach nearly 9 mmHg, exceeding the requirements of the AMMI standard. This further verifies the effectiveness of each design in our Efficient Feature Extraction mechanism.

7.5.3 Benefit of Channel-based Data Augmentation. In order to assess the contribution of ChannelDA, we remove the multi-channel PPW signals. We also change the loss function back to a single smoothL1Loss. As shown in the fifth row of Table 4, the main degradation is the STD that increases by 15%. The possible reason may be the models cannot be aware of the distortion in waveforms due to the inaccurate sensor position, as shown in Figure 6. To be specific, the pattern of distortion varies every time wearing it, which leads to unstable features. Our ChannelDA enforces the model to focus on the features immune to the sensor position by constructing a dataset containing the PPW in different positions and avoiding abnormal BP estimations.

7.6 Computational Cost

Although implemented on the powerful NVIDIA GeForce RTX 4090 GPU, PracticalBP still has the potential to be deployed on portable devices. Since the pretraining phase is not relevant to actual deployment, we only focus on the memory requirements and latency at the personalization and inference stages. We deploy the personalized model training and BP inference on an RTX 4090 card. As shown in Figure 5, the GPU memory requirement is 3832 MB and it needs 240 s to finish 100 epochs of training. And for BP inference, it only needs 74 ms to output a BP measurement for one 5-second input which indicates PracticalBP can work in real time and the required GPU memory is 2126 MB. Since the memory of the chip on the current commercial phone achieves 8192 MB [24], our model can be deployed practically.

8 RELATED WORK

8.1 Cuffless BP Measurement

Due to the discomfort caused by compression of blood vessels in the oscillometric BP measurement method, more and more cuffless BP measurement research has emerged in recent years. These studies can be mainly divided into two categories, wearable and contactless systems.

Wearable BP Measurement. Many physiological signals (e.g., ECG, PPG, and PPW) that can reflect cardiovascular information are used to infer BP in a comfortable way. In the beginning, researchers try to infer BP based on physiologically meaningful models. The most representative is the research based on PTT. This type of work is based on the Moens-Korteweg equation and Hughes Equation [40, 45] to model blood vessels and blood flow, and it is found that BP has a strong correlation with pulse wave transmission velocity. [11, 12, 54] obtain the PTT with ECG and PPG. But these works are on the basis of ideal assumptions and cannot achieve satisfactory accuracy. Then some works [9, 17, 19, 29, 35, 48, 50] propose the pulse wave analysis to extract more features from PPG and PPW to improve the performance. Haddad et al. [19] extract 27 features from pulse wave signals but the results cannot meet the requirements of AAMI. Benefiting from the development of deep learning, many studies [7, 36, 61, 65, 70, 74] have begun to use neural networks to allow the model to automatically find features related to BP. Li et al. [36] leverage a CNN to infer BP from multi-channel PPW. Crisp-BP [7] designs a BLSTM-based HNN to predict BP from PPG. Ji et al. [26] design a PPG-conditional Generative model-based framework to generate accurate ECG with PPG and apply a Bi-LSTM model to infer BP using the generated ECG and PPG signals. BP³ [74] introduces an attention module to fuse the pulse wave collected by a wearable mmWave and physiological information. However, these works either conduct the experiments under the LOSO setting only on small-scale datasets or do not meet AAMI standards. HearBP [72] proposes a cuffless BP monitoring system with heart sound collected by an earphone. HearBP cleans the heart sound using a U-net autoencoder-decoder and extracts the time domain and frequency domain features from phonocardiogram (PCG). But it needs the user's clean PCG to train a model for noise removal which makes it impractical.

Contactless Solution. Recently, some researchers have developed contactless methods to improve the convenience of BP measurement. CamBP [53] collects remote photoplethysmography (rPPG) from a regular webcam to infer BP by a neural network. But rppg is easily affected by ambient light and the introduction of cameras will lead to privacy issues. Huang et al. [22] capture BCG with a customized optical fiber sensor under the pillow and use a U²-Net to predict BP. But its performance in hypertensive degrades is unsatisfactory. Ishizaka et al. [25] apply LSTM to measure BP with pulse waves due to aortic vasomotion collected by a Doppler radar. But the requirement of dedicated hardware limits its usage. WaveBP [21] uses a commercial mmWave radar and senses the SCG from the chest. However, it needs a 20-minute calibration and mmWave radar is not a portable device.

Different from these works, PracticalBP is a wrist-worn device that is widely accepted by users and achieves satisfactory accuracy on a dataset containing normal and patients with hypertension in a user-friendly way.

8.2 Calibration-based Continuous BP Monitoring

With the explorations on cuffless BP measurement, researchers gradually realize that it is very difficult to accurately infer BP directly from physiological signals, and calibration becomes essential. To this end, some works try to use the user's actual BP labels and build user-specific models. PTT-based works need at least two BP measurements to determine the model parameters [4]. Lee et al. [32] inputs ECG, PPG, and BCG to a Bi-LSTM model, and 20% data of each user are used to finetune. Slapničar et al. [65] select 20% of a given subject's data to train a personalized Spectro-Temporal model as well. In addition, UTransBPNet [73] leverages 10% data of the given subject and achieves good accuracy in their own dataset which contains BP intervention experiments. However, such a large amount of personal data is difficult to obtain in practice. The system most similar to ours is FewShotBP [16]. FewShotBP devotes to reducing the calibration effort. However, it still needs 10 BP

measurements and complex BP intervention tasks to train a personalization adapter. And the calibration process will make the user feel discomfort. For example, the cold pressor test in FewshotBP needs the users to prepare the ice water and put their hands into it for one minute. In our experiment, one-third of volunteers report that they feel tingling in their hands during the cold pressor test. Moreover, frequent BP measurements may lead to arm soreness. In contrast, our system only needs one BP measurement and the BP intervention tasks are exempted.

9 DISCUSSION

9.1 Posture for BP Measurement

In the evaluation, all experiments are conducted by the subjects in the standard posture required by the cuff device. To be specific, the subject sits upright with their back supported, feet resting flat on the floor, and arms positioned at heart level [2]. The most critical point is that each subject needs to keep the relative height of our device and the heart from changing significantly because of the influence of hydrostatic pressure on the signals. Although this setting is acceptable during the day, it is always desired to achieve accurate BP measurement in other postures to expand usage scenarios, such as nighttime BP measurement. Nocturnal hypertension increases the risk of cardiovascular and all-cause mortality. In our future work, we plan to design a module that measures the height difference between the device and the heart to remove the posture restrictions.

9.2 The Complexity of Prototype

To collect high-quality PPW and PPG signals, we apply back pressure with a small airbag, a micro pump, and a micro valve. Although these additional modules increase the complexity and cost of the system, it's worth it because the modules can be reused to achieve oscillometric BP measurement. Recently, many commercial smart watches such as Omron BP8000 [1] and Huawei Watch D [23] have equipped a small airbag and micro pump to measure oscillometric BP on the wrist successfully. Therefore, we plan to further reuse the additional modules which eliminates the need for other cuff devices in our calibration process. Moreover, for the PPW sensing, we will try to use the air pressure sensor which is essential in oscillometric BP devices, and replace the piezoresistive sensor array. These will improve the practicality of our system and reduce the cost.

9.3 User Protocol in Real-World Scenario

In this work, the calibration process of the subjects is conducted under the guidance of a technician. In future practical use, to make sure the user finishes our calibration correctly, we plan to install a speaker on our wristband prototype and replace the participation of a human with a voice guidance. To be specific, the voice guidance will tell the user our calibration procedure step by step, such as when to tilt the arm. Meanwhile, the IMU will be used to monitor the user's arm tilt in real-time and guide the user to move the wrist to the appropriate height range. In addition, unconscious shaking of the user's arm or tremoring due to weak arm muscle strength will lead to low signal quality and degrade performance. To avoid it, the IMU will also be used to detect the user's motion artifacts and remind the user to perform the operation correctly through voice guidance. Considering that some users might find it hard to correctly complete part of some calibration data collection during the arm tilting process, our system will skip the step directly when three data collection failures are detected at each height. We have verified in Section 7.5.1 that the accuracy of PracticalBP still meets the requirements of the AAMI standard when some hydrostatic pressure-based pseudo labels are missing.

9.4 Valid Duration of Personalized Model

To understand the valid duration of our personalized model, we conduct a one-week experiment and verify that PracticalBP can ensure 7 days of performance with only a 4-minute calibration, which is affordable. In the future, we will further explore the maximum valid duration of the personalized model and the factors that affect the

validity time. In addition, since patients with abnormal BP need BP monitoring frequently, we will deploy our system to patients for long-term monitoring.

9.5 Abnormal BP Caused by Various Diseases

In this work, we only recruit 18 hypertension patients to demonstrate the effectiveness of our system. We are aware that further evaluating PracticalBP in patients who have abnormal BP caused by other diseases is meaningful. Although we do not study these populations in depth in this paper, some results show the potential of PracticalBP for BP monitoring in them. For instance, we find that the resting BP records of some subjects who are not diagnosed with hypotension in our dataset are in the hypotensive range (SBP/DBP is less than 90/60 mmHg). The characteristics of these subjects are mostly tall and thin or female. The PracticalBP's MAE of DBP and SBP in this case are 3.44 and 4.35 mmHg, respectively, which means that PracticalBP has the potential to measure the BP of hypotension. What's more, the type and dosage of medication are determined by the cause of the abnormal BP, and the effect of the drug use is also worth exploring. Since quantitative drug experiments should be operated under the supervision of doctors, we do not conduct them in this paper. However, many recruited hypertensive patients have the habit of taking antihypertensive drugs regularly. As shown in Figure 22, the patient who participated in our one-week experiment has the habit of taking medicine every morning, and our system successfully tracks the BP variations. At last, the main cause of the recruited participants' high BP is vascular aging. For other nonvascular hypertension, the cases that lead to hypertension are complex and diverse. To improve the scope of application of our system, we plan to cooperate with hospitals in the future and conduct more extensive experiments with the participation of doctors.

10 CONCLUSION

In this paper, we present PracticalBP, an accurate cuffless wearable BP monitoring system with user-friendly calibration. Compared with SOTA, we relieve the calibration effort by applying physiological principles to capture user-specific hemodynamic information. To achieve this, we leverage the principles of tonometry and hydrostatic pressure and propose a waveform-based channel selection algorithm and a discrete height measurement to generate a personalized dataset for each subject. Moreover, an effective feature extractor is designed for individual feature awareness and a channel-based data augmentation is used to improve the robustness. The evaluation results prove that our system meets the requirements of the AAMI standard and achieves a comparable performance with SOTA, while the calibration effort is reduced by a factor of five and the whole procedure is more convenient. We envision that PracticalBP can be used by people especially those with hypertension to monitor the risk of cardiovascular disease and guide the adjustment of drug dosage.

Acknowledgments

We are grateful to all the anonymous reviewers for their insightful comments. This research is supported in part by RGC under Contract CERG 16206122, 16204523, AoE/E-601/22-R, R6021-20, SRFS2425-6S05 and Contract R8015. This study is also funded in part by the Guangzhou Science and Technology Program (2024A03J1074, 2023A03J0286 and 2024A03J0927).

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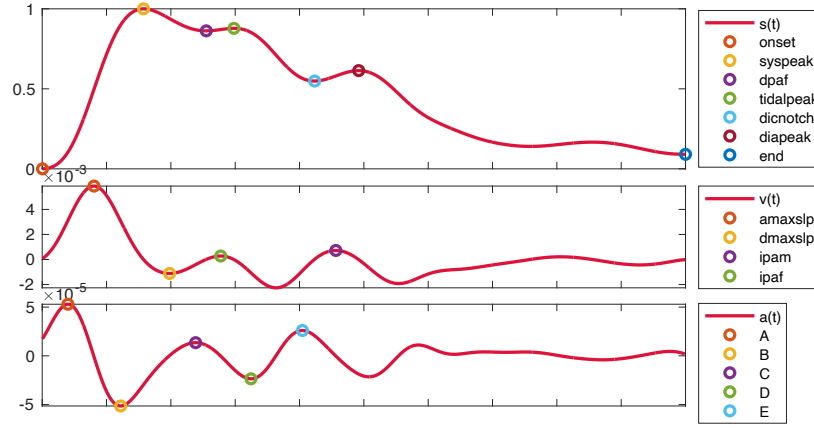


Fig. 24. The illustration of the manually extracted features.

Table 6. Definitions of manually extracted features.

Features	Description	Features	Description
AI_F	$s(t_{ipaf})/s(t_{syspeak})$	$RWTT_F$	$t_{ipaf} - t_{amaxslp}$
AI_M	$s(t_{ipam})/s(t_{syspeak})$	$RWTT_M$	$t_{ipam} - t_{amaxslp}$
$LASI_F$	$t_{ipaf}/t_{syspeak}$	$APULSE_A$	$a(t_A)$
$LASI_M$	$t_{ipam}/t_{syspeak}$	$APULSE_B$	$a(t_B)$
$IPA1$	$\int_{t_{onset}}^{t_{amaxslp}} s(t)dt$	$APULSE_C$	$a(t_C)$
$IPA2$	$\int_{t_{syspeak}}^{t_{onset}} s(t)dt$	$APULSE_D$	$a(t_D)$
$IPA3_F$	$\int_{t_{syspeak}}^{t_{ipaf}} s(t)dt$	$APULSE_E$	$a(t_E)$
$IPA3_M$	$\int_{t_{syspeak}}^{t_{ipam}} s(t)dt$	$RATIO_{BA}$	$s(t_B)/s(t_A)$
$IPA4_F$	$\int_{t_{ipaf}}^{t_{end}} s(t)dt$	$RATIO_{CA}$	$s(t_C)/s(t_A)$
$IPA4_M$	$\int_{t_{ipam}}^{t_{end}} s(t)dt$	$RATIO_{DA}$	$s(t_D)/s(t_A)$
$SDPTGDA$	$s(t_D)/s(t_A)$	$RATIO_{EA}$	$s(t_E)/s(t_A)$
$SDPTGBA$	$s(t_B)/s(t_A)$	AGI	$(s(t_B) - s(t_C) - s(t_D) - s(t_E))/s(t_A)$
$INFLECTION_F$	$v(t_{ipaf})$	APG	$(a(t_C) + a(t_D) - a(t_B))/a(t_A)$
$INFLECTION_M$	$v(t_{ipam})$		

A THE DETAILS OF MANUAL FEATURE EXTRACTION

To supplement the data-driven features and improve the effectiveness of BP-related feature extraction, we manually extract 27 features from each pulse in PPW and PPG respectively which are proven that have a high correlation with BP [14, 16]. As shown in Figure 24, these 27 features are extracted from the raw pulse $s(t)$, its first derivative $v(t)$, and its second derivative $a(t)$. The definitions of them are described in Table 6. Many of them have interpretable physiological significance. To be specific, the augmentation index (AI) and the velocity of the reflection point are the measure of the contribution of wave reflections to systolic arterial pressure and reflect changes in elastic arterial compliance. The large artery stiffness index (LASI) describes the transmission time of

the pressure wave from the root of the subclavian artery to the reflection point and then back to the subclavian artery, and is related to the stiffness of the large arteries. The IPA is the pulse area-related feature and it can be used as an indicator of total peripheral resistance. The reflected wave transit time (RWTT) is related to the time taken for the pressure wave to propagate from the heart to the periphery and back and it can be treated as a substitution of PTT. There are five important points in $a(t)$ which include a-wave (early systolic positive wave), b-wave (early systolic negative wave), c-wave (late systolic reincreasing wave), d-wave (late systolic redecaying wave) and e-wave (early diastolic positive wave). $RATIO_{BA}$ and $RATIO_{CA}$ reflect the increased and decreased arterial stiffness, respectively. $RATIO_{DA}$ is a useful index for the evaluation of vasoactive agents, as well as an index of left ventricular afterload. An increase of the $RATIO_{EA}$ reflects decreased arterial stiffness. Both AGI and APG are the indexes that indicate vascular aging and can be used to screen for arteriosclerotic disease. Note that due to the differences in each person's pulse waveform, not every feature can be successfully extracted. For example, as we discussed in Section 3 some participants' pulse waves do not have obvious tidal peaks. In these cases, features that are not successfully extracted are represented by missing. Our feature-wise attention mechanism sets different weights for different people's pulse features to eliminate these effects.